[INCH-POUND]
GGG-W-2843
May 7, 1993

FEDERAL SPECIFICATION
WRENCH, MICROMETER, TORQUE

This specification is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers micrometer adjustable, feel impulse, bi-directional, torque wrenches having either a non-ratchet or ratchet head. The torque wrenches are used to apply a pre-set torque to threaded fasteners and other torque requirements.

1.2 CLASSIFICATION

1.2.1 Classes. As specified (see 6.2), the torque wrench shall be one of the following classes:

Class 1. Non-ratchet head (precise clockwise torquing)
Class 2. Ratchet head (precise clockwise torquing)
Class 3. Non-ratchet head (precise counter-clockwise torquing)
Class 4. Ratchet head (precise counter-clockwise torquing)

1.2.2 Class accuracy. Class 1 and Class 2 torque wrenches have an accuracy of ± 4 percent in the clockwise direction and ± 6 percent in the counter-clockwise direction of applied torque. Class 3 and Class 4 torque wrenches have an accuracy of ± 4 percent in the counter-clockwise direction and ± 6 percent in the clockwise direction of applied torque.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. The following documents, of the issue in effect on the date of invitation for bid, or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

QQ-F-838 - Fusible Alloys, Ingots and Bars
PPP-P-40 - Packaging and Packing of Hand Tools

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FSC 5120

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Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-45662 - Calibration Systems Requirements

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply:

American National Standards Institute:

ANSI B107.4 - Driving and Spindle Ends for Portable Hand, Air, and Electric Tools

ANSI B107.10M - Handles and Attachments for Hand Socket Wrenches - Inch and Metric Series

(Application for copes should be addressed to The American National Standards Institute, 1430 Broadway, New York, NY 10018.)

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American Society for Testing and Materials:

E 18-84 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials, Standard Methods of Test for

ASTM B 487 - Standard Method for Measurement of Metal and Oxide Coating Thickness by Microscopical Examination of a Cross Section

ASTM B 499 - Standard Test Method for Measurement of Coating Thickness by the Magnetic Method: Nonmagnetic Coatings of Magnetic Basis Materials

ASTM B 504 - Standard Test Method for Measurement of Coating Thickness by the Coulormetric Method

ASTM B 530 - Standard Test Method for Measurement of Coating Thickness by the Magnetic Method: Electrodeposited Nickel Coatings on Magnetic and Nonmagnetic Substrates

ASTM B 568 - Standard Test Method for Measurement of Coating Thickness by X-Ray Spectrometry

ASTM B 748 - Standard Method for Measurement of Thickness of Metallic Coatings by Measurement of Cross Section with a Scanning Electron Microscope

ASTM A 754 - Standard Test Method for Coating Thickness by X-Ray Fluorescence

(Application for copes should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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3. REQUIREMENTS

3.1 Torque wrench. The torque wrench shall be bi-directional and shall have an accurate micrometer-type, pre-setting mechanism upon which the required torque can be hand set by adjusting and locking the mechanism without the use of special tools or equipment. The torque wrench shall have a positive locking device which holds the pre-set torque securely during operation of the wrench. When the applied torque equals the pre-set torque, the torque wrench shall momentarily release and signal that the pre-set torque has been reached by feel impulse with or without audible signal. When hand pressure is removed from the torque wrench handle, the torque wrench shall automatically reset for the next operation.

3.2 Illustrations. The illustrations herein are descriptive and are not intended to preclude the purchase of torque wrenches otherwise conforming to this specification.

3.2.1 Torque wrench - Class 1 and 3. Class 1 and 3 torque wrenches shall be similar to figure 1; except 500-2000 foot-pound capacity torque wrenches shall be similar to figure 2.

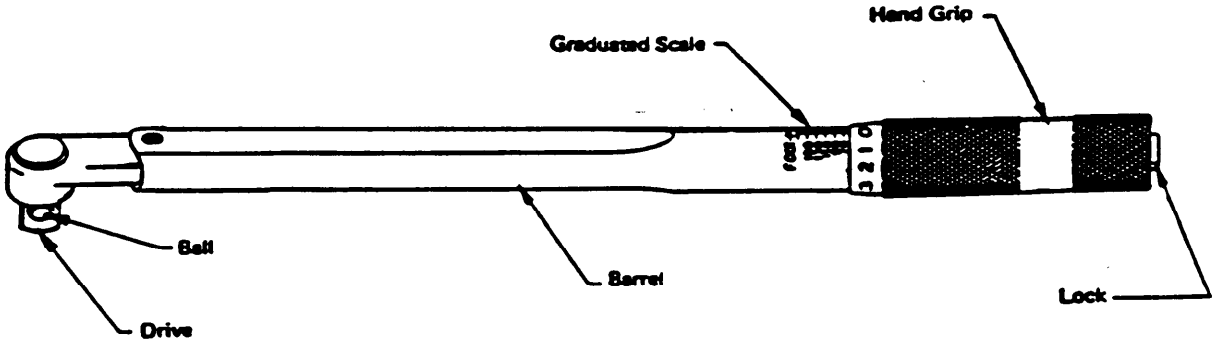


FIGURE 1. Class 1 and 3 torque wrench, micrometer adjustable, torque pre-setting, feel impulse, non-ratcheting head.

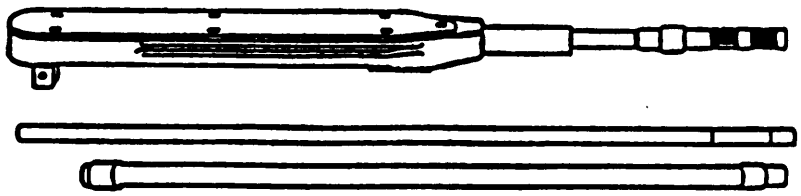


FIGURE 2. Class 1 and 3 torque wrench (500-2000 foot-pound capacity), micrometer adjustable, torque pre-setting, feel impulse, non-ratcheting head.

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3.2.2 Torque wrench - Class 2 and 4. Class 2 and 4 torque wrenches shall be similar to figure 3.

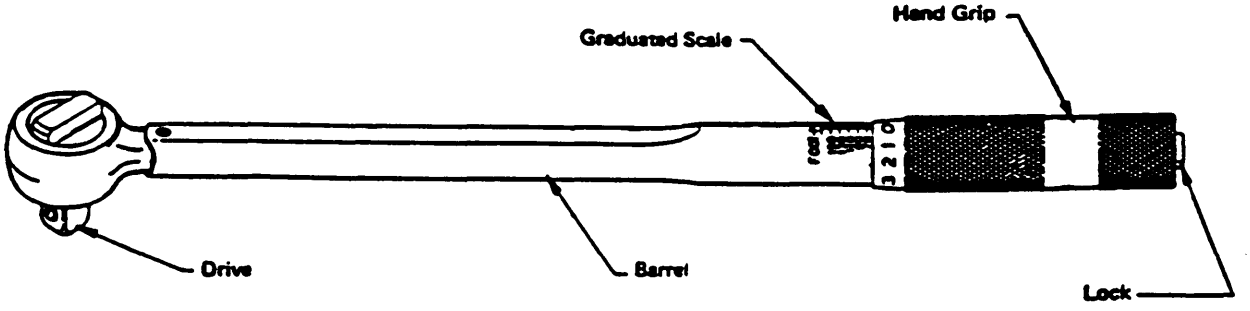


FIGURE 3. Class 2 and 4 torque wrench, micrometer adjustable, torque pre-setting, feel impulse, ratcheting head.

3.3 First article samples. Unless otherwise specified (see 6.2), two torque wrenches of each kind (class, rated torque capacity, drive size, graduation increments) shall be submitted for testing in accordance with the requirements of 4.2, 4.4, and 4.5.

3.4 Design, materials, and construction. The torque wrench design, materials (e.g., composition, hardness, plating), and construction shall be of uniform high quality adequate to meet the requirements of this specification and capable of providing repeated accuracy and extended service.

3.4.1 Handle. The torque wrench shall have a metal handle of one-piece or multi-piece design using a threaded or other mechanically fitted joint. A suitable portion of the handle shall be knurled, grooved, ribbed, or provided with a non-toxic rubber cover that facilitates turning of the adjustable portion of the handle.

3.4.2 Drive requirements. The torque wrench shall be provided with a male square drive with a spring loaded steel ball and shall conform to the requirements of ANSI B107.4. The spring loaded steel ball detent shall be located as shown in figure 1. The male square drive hardness shall be Rockwell C 34 to 54. The male square drive shall not be welded or made of powdered metal. On non-ratcheting torque wrenches, two opposite flats of the male square drive shall be parallel within 5 degrees of the longitudinal axis of the torque wrench handle.

3.4.3 Ratchet requirements. Torque wrenches containing a ratchet in the torque wrench head shall be in accordance with the applicable requirements of ANSI B107.10M.

3.4.4 Torque measuring elements. The torque measuring elements used in the torque wrench shall be of uniform high quality and capable of providing repeated accuracy and extended service.

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3.4.5 Torque setting mechanism. The torque wrench shall incorporate a locking device which prevents the pre-set torque value from accidently changing while the wrench is being used. Setting and locking the pre-set torque value shall be accomplished without the use of tools or equipment. If the torque setting mechanism is rotated one revolution below the minimum rated torque capacity specified in Table I, uncoupling of the mechanism shall not occur.

3.4.5.1 Torque calibration/recalibration adjustment mechanism. The torque wrench shall incorporate an internal mechanism which can be adjusted externally to optimize the wrench accuracy during calibration and recalibration. Subsequent to calibration and recalibration, the internal adjustment mechanism shall be capable of being externally sealed in a manner that prevents tampering with the adjustment using a sealing alloy that conforms to QQ-F-838, Composition A.

3.4.6 Signal and release mechanism. The torque wrench shall incorporate a torque limiting device which momentarily and automatically releases and signals by feel impulse with or without audible signal when the pre-set torque value has been reached.

3.5 Torque scale. The torque wrench shall have a graduated torque scale in the applicable units of torque measurement from the minimum to the maximum rated torque capacity specified in Table I. The torque scale shall be divided into equal graduation increments with each increment's size equal to or less than specified in Table I. Arabic numbers shall be at intervals not greater than each tenth graduation line. The graduation lines and numbers shall be cut, etched, or stamped in a manner which causes permanent deformation of the metal without damage to the finish coatings. The torque wrench shall be permanently marked on the top of the torque wrench with the units of torque measurement (e.g., ft-lbs, in-lbs) and the word "TORQUE" or "TORQUES" with an arrow indicating the $\pm$ 4 percent calibrated torque direction (see Table II). All markings shall be permanent, legible, and of adequate size and contrast for reading in both daylight and artificial light.

3.5.1 Extended torque scale markings. If the torque scale extends below the minimum or above the maximum rated torque capacity specified in Table I, the extended scale markings shall be consistent with the scale markings over the rated torque capacity and in conformance with the requirements of 3.5 over the extended torque capacity (see 3.6.1).

3.6 Torque capacity. Unless otherwise specified (see 6.2), the torque wrench shall have, as a minimum, the applicable torque rated capacity specified in Table I. Unless otherwise specified (see 6.2), the torque wrench may have an extended torque capacity (see 3.6.1) beyond that specified. Torque wrenches less than 600 foot-pound capacity shall be of such length that a maximum force of 150 pounds (when applied to the center of the handgrip or to a handle extender) shall be sufficient to apply the maximum rated torque without changing the specified accuracy.

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TABLE I. Micrometer torque wrench

RATED TORQUE CAPACITY (MINIMUM - MAXIMUM)		DRIVE SIZE	INCREMENT GRADUATIONS (MAXIMUM)	CLASS			
Inch-pounds	Foot-pounds	Inch		1	2	3	4
10-50	--------	1/4	1 in.-lb.	X	X	X	X
30-150	--------	1/4	2 in.-lbs.	X	X	X	X
30-150	--------	3/8	1 in.-lb.	X	X	X	X
40-200	--------	1/4	1 in.-lb.	X	X	X	X
40-200	--------	1/4	2 in.-lbs.	X	X	X	X
40-200	--------	3/8	1 in.-lb.	X	X	X	X
150-750	--------	3/8	5 in.-lbs.	X	X	X	X
200-1000	--------	3/8	5 in.-lbs.	X	X	X	X
360-1800	--------	1/2	10 in.-lbs.	X	X	X	X
500-2500	--------	1/2	10 in.-lbs.	X	X	X	X
600-3000	--------	1/2	10 in.-lbs.	X	X	X	X
700-1600	--------	1/2	10 in.-lbs.	X	X	X	X
1200-4800	--------	3/4	40 in.-lbs.	X	X	X	X
---------	15-75	3/8	1/2 ft.-lb.	X	X	X	X
---------	16-80	3/8	1/2 ft.-lb.	X	X	X	X
---------	16-80	1/2	1/2 ft.-lb.	X	X	X	X
---------	20-100	3/8	1/2 ft.-lb.	X	X	X	X
---------	30-150	1/2	1 ft.-lb.	X	X	X	X
---------	40-200	1/2	1 ft.-lb.	X	X	X	X
---------	50-250	1/2	1 ft.-lb.	X	X	X	X
---------	60-300	3/4	5 ft.-lbs.	X	X	X	X
---------	100-500	3/4	5 ft.-lbs.	X	X	X	X
---------	120-600	3/4	5 ft.-lbs.	X	X	X	X
---------	140-700	1	2 ft.-lbs.	X	X	X	X
---------	500-2000	1	10 ft.-lbs	X		X	

x = Available in this class.

3.6.1 Extended torque capacity. If the torque wrench has an extended torque capacity with applicable torque scale markings below the minimum or above the maximum rated torque capacity specified in Table I, the torque wrench shall meet all requirements of this specification, as modified (see 3.6.2), over the extended torque capacity.

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3.6.2 Extended torque capacity specifications. If the torque wrench has an extended torque capacity (see 3.6.1), the specified requirements in this specification shall be modified as follows: Substitute the lowest torque value marked on the torque wrench in lieu of the minimum rated torque capacity. Substitute the highest torque value marked on the torque wrench in lieu of the maximum rated torque capacity. Substitute extended torque capacity in lieu of rated torque capacity.

3.7 Torque overload. The torque wrench shall be capable of repeated overloads of 125 percent of the maximum rated torque capacity in both directions without damage, deformation, or loss of accuracy over the rated torque capacity. Subsequent to overload, the torque wrench shall, without repair or recalibration, be capable of meeting the requirements of this specification.

3.8 Torque accuracy and direction.

3.8.1 Performance. Unless otherwise specified (see 6.2), the torque wrench shall indicate the applied torque over the rated torque capacity specified in Table I within the accuracy tolerances and directions specified in Table II. All specified tolerance limits are absolute limits.

TABLE II. Torque Wrench Accuracy Tolerances.

Accuracy tolerance		
Class	Clockwise Direction	Counter-Clockwise Direction
1 & 2	± 4 percent	± 6 percent
3 & 4	± 6 percent	± 4 percent

3.8.2 Pre-conditioning. Immediately prior to commencing testing, the torque wrench shall be pre-conditioned at 50 percent of the maximum rated torque capacity for three excitations in each torque direction in which the testing is to be performed (see 4.5.1).

3.8.3 Calibration accuracy. The torque wrench shall be calibrated within the accuracy requirements specified in 3.8.1. The calibration shall be performed in accordance with the test requirements specified in 4.8 using test equipment that complies with the requirements of 4.8.2. Subsequent to calibration, the torque wrench shall be protected against tampering by sealing the adjustment mechanism with a sealing alloy that conforms to QQ-F-838, Composition A. Subsequent to sealing of the adjustment mechanism, the torque wrench shall meet the accuracy requirements specified in 3.8.1.

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3.8.4 Recalibration accuracy. Without repair, a torque wrench which has not been abused or misused shall be capable of being recalibrated within the accuracy requirements specified in 3.8.1 during an extended service period. The extended service period shall be limited to a total of 16,000 operational cycles in the precision torquing direction and 4,000 operational cycles in the non-precision torquing direction. The operational cycles shall have an approximate normal distribution over the rated torque capacity. Recalibration shall be performed in accordance with the test requirements specified in 4.8 using test equipment that complies with the requirements of 4.8.2. Recalibrated torque wrenches shall maintain calibration for a minimum of 90 days of combined storage and operational use. The operational use shall be limited to 4,000 cycles in the precision torquing direction and 1,000 cycles in the non-precision direction. If the torque wrench has not been operated for 90 days or more, the wrench need not initially meet the accuracy requirements of 3.8.1; but, shall be within $\pm$ 6 percent accuracy in both directions and shall be capable, without repair, of being recalibrated within the accuracy requirements of 3.8.1 during the extended service period.

3.8.5 Endurance accuracy. The torque wrench shall maintain the accuracy specified in 3.8.1 for 5,000 cycles in each direction of operation at the maximum rated capacity without being deformed, damaged, requiring repair, or recalibration. Subsequently, the torque wrench shall be capable of 20,000 cycles in each direction of operation at 50 percent of maximum rated torque capacity without being deformed, damaged, or requiring repair; but may, if required, be recalibrated prior to accuracy testing, and shall meet the accuracy requirements specified in 3.8.1.

3.8.6 Positional sensitivity accuracy. The torque wrench shall indicate the applied torque over the rated torque capacity specified in Table I within the accuracy requirements specified in 3.8.1 with any orientation of the torque wrench with respect to gravity.

3.9 Finish and coatings. Cadmium plating shall not be used as an external or internal protective coating. Torque wrench internal parts subject to corrosion shall be protected by the use of lubricants or other protective coatings. Torque wrench external ferrous parts shall be smoothly finished and protected to prevent corrosion. The male tang, spring loaded steel ball, screws and other fasteners shall be protected by a corrosion resistant coating. The corrosion resistant coating shall be in addition to any lubricant(s). All other external ferrous parts shall be protected with a chromium plated coating (see 3.9.1).

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3.9.1 Chromium plated coating. The torque wrench shall be electroplated with a nickel (Ni) coating followed by a chromium (Cr) coating. There shall be no other coatings or undercoatings. The coating minimum thickness shall be 0.000200 inch or greater for nickel and 0.000007 inch or greater for chrome. Visible contact marks resulting from the electroplating operations shall be confined to the interior of the torque wrench. Surfaces of the electroplated torque wrench shall be adherent, smooth, continuous, and free of visible defects (e.g., blisters, pits, roughness) which might adversely affect the serviceability, durability, or safety of the torque wrench.

3.10 Marking. Unless otherwise specified (see 6.2), the torque wrench shall be permanently marked with the manufacturer's name or identifying symbol, the country of origin, and the month and year of manufacture. If the torque wrench is marked with a manufacturer assigned unique serial number, the month and year of manufacture is not required. Optional markings (e.g., serial number, model number) which do not interfere with or degrade the use of the torque wrench may be included. All markings shall be cut, etched, or stamped in a manner which causes permanent deformation of the metal without damage to the finish and coatings (see 3.9). The markings shall be legible, permanent, and of adequate size and contrast for reading in both daylight and artificial light.

3.11 Carrying case. Unless otherwise specified (see 6.2), each torque wrench shall be provided in a carrying case. The carrying case shall be either blow molded plastic or metal. The case shall be provided with a catch which cannot be opened without disengagement of a positive latch. The case interior shall be close fitting to accommodate the torque wrench when set at the minimum rated torque capacity and shall limit movement during shipment, handling, and storage. The case shall be of sturdy construction sufficient to protect the torque wrench and accessories, if any, from damage during shipment, handling, and storage.

3.11.1 Carrying case marking or label. The outside of the top lid of the carrying case shall be permanently marked or labeled (label shall remain permanently affixed to the carrying case for the extended service period of the torque wrench) with the following applicable information:

Torque capacity:
Accuracy clockwise:
Accuracy counter-clockwise:
Drive size:
Graduation increments:
National Stock Number:
Specification:
Month and year of manufacture or serial number:
Contract or Purchase Order Number:

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3.12 Instructions. Unless otherwise specified (see 6.2), each torque wrench shall be packaged with a set of instructions which cover the application, operation, calibration, and maintenance of the torque wrench, adapters, and handle extenders. The instructions shall identify the specified clockwise and counter-clockwise torque wrench accuracy and shall state that the accuracy is applicable over the torque wrench scale. If applicable, warranty information shall be provided with the instructions.

3.13 Calibration record. Each torque wrench shall be packaged with a calibration record which delineates the manufacturer's name; month and year of manufacture or manufacturer assigned unique serial number, contract or purchase order number; torque capacity and specified accuracy tolerances and torque directions; date of calibration; and the actual calibration readings (see 3.8.3) applicable to the torque wrench.

3.14 Workmanship. Workmanship shall be of the highest grade throughout and equal in every respect to the best commercial practices. The torque wrench shall be free of all defects that would impair serviceability, durability, safety or appearance.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection of material and components. The supplier is responsible for insuring that materials and components used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified herein, or, if none, in accordance with this specification.

4.2 First article. Unless otherwise specified (see 6.2), two torque wrenches of each kind (class, rated torque capacity, drive size, graduation increments) shall be examined in accordance with 4.4 and tested in accordance with 4.5. Failure of any examination or test shall be cause for rejection.

4.3 Lot acceptance. The Government reserves the right to have a Government representative select samples from random lots during the contract period and send them to a Government or independent laboratory for testing.

4.3.1 Lot sampling procedures. Sampling procedures for torque wrench lots shall be in accordance with MIL-STD-105. Data for sampling shall be as specified in Table III.

4.3.2 Lot acceptance performance test. For lot acceptance, the sample torque wrench shall be tested a total of five (5) times in accordance with the Accuracy Test requirements (see 4.8) and the Accuracy Test Procedure (see 4.8.1).

4.3.3 Lot acceptance endurance test. For lot acceptance, the sample torque wrench shall be tested in accordance with the Endurance Test requirements (see 4.6.3) and Endurance Test Procedure (see 4.6.3.1) with the following exceptions: (a) The total number of cycles for each wrench shall be 2,000 times in each direction from a no-load condition to the maximum rated torque capacity; (b) The requirement to cycle 20,000 times in each direction from a no-load condition to 50 percent of the maximum rated torque capacity is not applicable; (c) the number of samples tested shall be as determined by the lot sampling procedures (see 4.3.1).

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TABLE III. Sampling data.

Category	Sample Unit	Inspection level	Acceptable quality level	AQL in terms of	Reference
Visual examination	1 ea.	S-4	Major 2.5	Defects per hundred units	4.4.1
Dimensional examination	1 ea.	S-3	Major 2.5	Defects per hundred units	4.4.2
Testing	1 ea.	S-2	4.0	Percent defective	4.3.2 and 4.3.3
Testing (except as specified)	1 ea.	S-2	4.0	Percent defective	4.7 thru 4.11
Inspection for delivery	1 ea.	S-2	4.0	Percent defective	4.12

4.4 Examinations.

4.4.1 Visual examination. The sample torque wrench shall be examined for any nonconformance in design, material, construction, finish, coating, construction, marking, and workmanship. Defects are listed in Table IV.

4.4.2 Dimensional examination. The sample torque wrench shall be examined for any nonconformance with dimensional requirements.

4.5 Testing. The sample torque shall be tested in accordance with the requirements of 4.6 through 4.11.

4.5.1 Pre-conditioning. The sample torque wrench shall be pre-conditioned (see 3.8.2) immediately prior to performance of the Lot Acceptance Performance Test (see 4.3.2), the Lot Acceptance Endurance Test (see 4.3.3), the First Article Performance Test (see 4.6.1), the First Article Endurance Test (see 4.6.2), the First Article Positional Sensitivity Test (see 4.6.3), the Overload Test (see 4.7), and the Accuracy Test (see 4.8).

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TABLE IV. Classification of defects.

Categories	Defects
Critical:	
1	None defined.
Major:	
101	Class, rated torque capacity, drive size, graduation increments, not as specified.
102	Materials or construction not as defective required or specified.
103	Scale not as specified; illegible, graduations not uniform, not adequately spaced, not permanent, or graduations not identified (numbered) as required.
104	Male tang not as specified; not provided with a spring-loaded ball, ball not staked in place or improperly located.
105	Signal mechanism utilized for producing the torque indication not as specified.
106	Torque setting mechanism not as required or specified; cannot be readily adjusted or setting cannot be locked.
107	Ratcheting mechanism defective or not as specified (as applicable).
108	Adapters, extenders not provided (where applicable).
109	Marking not as specified; illegible, incorrect, or not permanent.
110	Carrying case not provided or as specified.
111	Instructions and calibration record not provided or as specified.
112	Workmanship not as specified; evidence of cracks, deep pits, heavy scale, deep tool marks, porosity, or foreign inclusions, not smoothly finished, fins, burrs, rough surfaces or sharp edges.

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4.6 First article testing. Failure during any test shall be cause for rejection.

4.6.1 First article performance test. The sample torque wrench shall be tested a total of twenty-five (25) times in accordance with the Accuracy Test requirements (see 4.8) and the Accuracy Test Procedure (see 4.8.1). There shall be no more than five complete tests in any 24 hour period.

4.6.2 First article endurance test. Unless otherwise specified (see 6.2), the First Article Endurance Test shall be performed on only one sample of the torque wrench. The test shall be performed in accordance with 4.6.2.1 and shall consist of 5,000 cycles in each direction performed at maximum rated torque capacity followed by 20,000 cycles in each direction performed at 50 percent of maximum rated torque capacity.

4.6.2.1 Endurance test procedure. The sample torque wrench shall be pre-set at the maximum rated torque capacity. The sample torque wrench shall be cycled 5,000 times in one direction from a no-load condition to the maximum rated torque capacity. The sample torque wrench shall be cycled continuously and at a rate not greater than 60 times per minute. Torque loading requirements (see 4.8.2.1) are not applicable. At the conclusion of the 5,000 cycle test, the sample torque wrench shall not be deformed, damaged, or require repair and shall not require recalibration. The torque wrench shall meet the accuracy requirements in accordance with the Accuracy Test (see 4.8). The above 5,000 cycle test procedure and requirements shall be repeated in the opposite direction. Using the above procedure, the torque wrench shall then be cycled 20,000 times in each direction from a no-load condition to 50 percent of the maximum rated torque capacity. At the conclusion of the 20,000 cycles in each direction, the sample torque wrench shall not be deformed, damaged, or require repair; but may, if required, be recalibrated prior to meeting the accuracy requirements. The torque wrench shall meet the accuracy requirements in accordance with the Accuracy Test (see 4.8).

4.6.3 First article positional sensitivity test. The positional sensitivity test shall be performed in accordance with 4.6.3.1.

4.6.3.1 Positional sensitivity test procedure. The sample torque wrench shall be vertically positioned 90 ± 5 degrees from the horizontal. The sample torque wrench handle's initial position shall be horizontal with the torque wrench head to the left. The torque wrench shall be set to the minimum rated torque capacity. Torque shall then be applied three times in both the clockwise and counter-clockwise directions. The torque shall be applied smoothly and perpendicular to the axis of the drive tang. At the instant the signal mechanism is actuated, the tester value shall be recorded. The sample torque wrench's handle shall then be positioned in the opposite horizontal position with the torque wrench head to the right, and the above procedure repeated. Each of the twelve (12) recorded values shall be within the requirements specified in 3.8.1.

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a. Restore equipment configuration

4.7 Overload test. The overload test shall be performed by loading the torque wrench to 125 percent of the maximum rated torque capacity three times in each direction. The torque wrench shall not be damaged, deformed, or require repair; and shall without recalibration, be tested in accordance with the Accuracy Test (see 4.8).

4.8 Accuracy test. The accuracy test shall in accordance with the requirements of 3.8 and shall be performed in accordance with 4.8.1.

4.8.1 Accuracy test procedure. The torque wrench shall be set to the minimum rated torque capacity specified in Table I and secured to the tester in a horizontal position. Torque load shall then be smoothly applied perpendicular to the axis of the drive tang. At the instant the torque wrench signal mechanism is actuated, the tester value shall be recorded. This procedure shall be repeated two additional times for a total of three readings at the minimum rated torque capacity in one torque direction. Without changing the setting of the torque wrench, the procedure shall be repeated in the opposite direction. The torque wrench shall then be adjusted to 60 percent of the maximum rated torque capacity specified in Table I and the procedure repeated in both directions. The torque wrench shall then be adjusted to the maximum rated torque capacity specified in Table I and the procedure repeated in both directions. Each of the eighteen (18) readings shall be within the accuracy requirements specified in 3.8.1.

4.8.2 Test equipment. Testing shall be performed using test equipment capable of measuring and displaying the first peak of applied torque within an accuracy of ± 0.5 percent or better at any point within the rated torque capacity of the torque wrench. Accuracy of test equipment elements (e.g., test weights, bars) which affect test measurement accuracy shall be traceable to the National Institute of Standards and Technology in accordance with the MIL-STD-45662. Interpolated readings of the test measuring equipment shall be made no more precisely than one-half of the smallest increment.

4.8.2.1 Torque loading. During the initial 80 percent of the pre-set torque value, the rate of torque loading is optional. During the final 20 percent of the pre-set torque value, the rate of torque loading shall not exceed 5 percent per second of the pre-set torque value. Torque loading shall be applied at the center of the handle or handle extender (if applicable). Except for the cyclic portion of the Endurance Tests (see 4.3.3 and 4.6.2) and the Overload Test (see 4.7), the torque loading requirements shall be applicable to all test requirements which utilize the test equipment.

4.9 Drive test. The sample torque wrench drive shall be measured and tested to establish conformance with the requirements of 3.4.2

4.9.1 Drive square hardness test. The sample torque wrench shall be subjected to a hardness test to verify conformance to the male drive square hardness requirements specified in 3.4.2. The test shall be preformed in accordance with ASTM E 18-84.

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4.10 Ratcheting mechanism test. The ratcheting mechanism shall be measured and tested to established conformance with the requirements of 3.4.3.

4.11 Chromium plating thickness test. At a randomly selected position on the torque wrench, measurement of each electroplated coating shall be conducted in accordance with any of the following, as appropriate, methods: ASTM B 487, ASTM B 499, ASTM B 504, ASTM B 568, ASTM B 748, or ASTM A 754.

4.11.1 Chromium plating adhesion test. At a randomly selected position on the torque wrench, a test for adhesion of electroplated materials shall be conducted in accordance with ASTM B 571 by one of the following methods: Grind-Saw, Push, File, or Burnishing.

4.12 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the preservation, packaging, packing, and marking comply with the requirements of this specification (see section 5.).

5. PREPARATION FOR DELIVERY

5.1 Preparation, packaging, packing, and marking. Preservation, packaging, packing, and marking shall be in accordance with PPP-P-40. The level of preservation and packaging shall be A or C, and the level of packing shall be A, B, or C, as specified (see 6.2).

6. NOTES

6.1 Intended use. Torque wrenches are used to apply torque to fasteners (e.g., bolts, nuts) and other torque requirements.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.

(b) Torque class (see 1.2), rated torque capacity, drive size, increment graduations (see Table I), required.

(c) If wrenches of a different torque capacity, drive size, increment graduations (see Table I), or accuracy (see Table II) are required.

(d) If First Article samples are not required (see 3.3).

(e) If a rubber covering over the adjustable handle is required (see 3.4.1).

(f) If an extended torque capacity is not acceptable (see 3.6).

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(g) If carrying case is not required (see 3.11).

(h) If instructions are not required (see 3.12).

(i) If a calibration record is not required (see 3.13)

(j) If more than one First Article sample is to be endurance tested (see 4.4.6).

(k) If different than specified, the level of preservation, packaging, and packing required (see 5.1).

MILITARY INTEREST

Military Coordinating Activity
Army - GL

Custodians
Air Force - 99
Navy - SH

Review Activity
Air Force - 30

User Activity
Navy - MC

PREPARING ACTIVITY

GSA - FSS (6FETE-CO)

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